Total No. of Questions : 6]

SEAT No. :

P5192

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B.E./Insem.-593
B.E. (Information Technology) (Semester-I)
MACHINE LEARNING
(2012 Pattern)

Time : 1 Hour] *[Maximum Marks : 30*

Instructions to the candidates:

1) *Draw neat diagrams wherever necessary.*
2) *Assume suitable data, if necessary.*
3) *Figures to the right indicate full marks.*

Q1) a) Explain supervised learning and unsupervised learning. **[5]**

b) Explain geometric models. **[5]**

OR

Q2) a) With an example, explain feature as a split and feature as a predictor. **[5]**

b) Explain grouping and grading models. **[5]**

Q3) a) Calculate true negative rate (tnr), accuracy and pos for the following: **[5]**

	Predicted+	Predicted-
Actual+	50	25
Actual–	5	20

b) Explain VC dimension. **[5]**

OR

Q4) a) Prove: precision = TP/(TP+FP) **[5]**

b) Explain overfitting and underfitting. **[5]**

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Q5) a) What is lasso and ridge regression? **[5]**

b) Explain perceptron. **[5]**

OR

Q6) a) Explain slack variables, hard margin and soft margin. **[5]**

b) Explain sigmoid function. **[5]**

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